

THARUN SUBRAMANYA

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EDUCATION

Master of Science – Artificial Intelligence, University of Michigan–Dearborn | GPA: 3.74/4.00

May 2026

Relevant coursework: Data Mining, Pattern Recognition and Neural Networks, Deep Learning, Automotive Sensors and Actuators, Edge Computing.

Post Graduate Certificate – Artificial Intelligence Specialization, Vellore Institute of Technology | CGPA: 8.46/10

Jun 2024

Relevant coursework: Statistics and Probability, Python for Data Analytics, Data Visualization, Big Data Analytics.

Bachelor of Engineering – Industrial and Production Engineering, The National Institute of Engineering | CGPA: 7.45/10

Sep 2022

Relevant coursework: Statistical Quality Control (statistical process control, hypothesis testing), Operations Research, Applied Mathematics (Calculus, Linear Algebra, Numerical Methods), Metrology and Measurements.

SKILLS

Programming Languages: Python, SQL, Spark SQL, R

Machine Learning Frameworks: scikit-learn, XGBoost, PyTorch, TensorFlow/Keras, pandas, NumPy, SHAP

AI, NLP & LLM Frameworks: NLTK, Hugging Face Transformers, LangChain, LlamaIndex

Data Engineering: PySpark, Apache Spark, MQTT, schema validation

Visualization & Statistical Graphics: Power BI, Tableau, Grafana, Plotly, matplotlib, seaborn

Cloud & Databases: AWS, Azure, PostgreSQL, MySQL, InfluxDB, SQLite

DevOps & Development Tools: Docker, Git, Linux, pytest, CI/CD, FastAPI, Flask, REST APIs

RESEARCH EXPERIENCE

Volunteer Research Assistant – Earmold 3D Model Segmentation for Machine Learning

Ongoing

- Annotate anatomical landmarks and regions on **3D ear impression scans (STL meshes)** across left and right ears, colour-shading regions in **collaboration** with the supervising researcher.
- Construct a **graph representation of the mesh** – vertices as nodes, connectivity as edges – as input for **graph neural network (GNN) 3D geometry segmentation**, then visualize the graph to confirm the conversion.
- Establish a **reproducible preprocessing path** from raw **unstructured** scan geometry to model-ready inputs, **communicating** labeling rules, assumptions, and limitations to research **stakeholders** in written **documentation**.

PROJECTS

Chappie – Retrieval-Augmented Assistant over an Unstructured Text Corpus

Ongoing

Python, FastAPI, Qdrant, Hugging Face/ONNX, Groq, pytest, Docker, GitHub Actions

- Built a **retrieval-augmented generation** assistant mining an **unstructured text** corpus behind a **FastAPI service** with guardrails.
- Engineered **hybrid retrieval** – BM25 lexical scoring fused with dense vector search by reciprocal rank fusion, then **cross-encoder reranking** – inside a 512 MB memory budget.
- Wrote a **97-case evaluation suite** gating deployment in continuous integration at **0% prompt-injection success**.

CANalyse Edge – Vehicle Fault Diagnostics and Data-Quality Validation Service

Aug 2026

Python, python-can, cantools, scikit-learn, FastAPI, pytest, GitHub Actions

- Designed a **CAN/CAN-FD diagnostic pipeline** decoding raw frames into named engineering signals through **DBC mapping**, recording timing, data-length code (DLC), and signal-range checks as **data-quality evidence**.
- Exposed results through a **Service-Oriented Vehicle Diagnostics (SOVD)**-inspired dashboard supporting **issue triage and root cause analysis**, evaluated under a **group-level sampling design** holding out unseen sessions.
- Implemented **maintainable test automation** – a repeatable pytest regression suite in **documented** continuous integration.

Supply Chain Risk Analytics – Large-Scale Signal Detection Pipeline

Jul 2026

Python, pandas, NumPy, scikit-learn

- Built a **repeatable, reusable data pipeline** over **180,000+ structured records** across **53 attributes**: cleaning, **schema validation**, and **predictive modeling** of late-delivery trends.
- Engineered temporal, operational, and **categorical data** features across a **multivariate** space with **leakage-safe preprocessing**, cutting one-hot dimensionality from **64,000+ features to 339**.
- Applied **statistics to calibrate the decision threshold** against precision, recall, F1, and ROC-AUC, lifting late-risk recall from **57.5% to 78.8%** under a cost-asymmetric objective.

DARE-PM – Anomaly Detection, Drift Diagnosis, and Prognostics on Sensor Streams

Apr 2026

Python, scikit-learn, One-Class SVM, ADWIN, MQTT, Docker, InfluxDB, Grafana

- Applied **One-Class SVM anomaly and outlier detection to time series** sensor streams fitted on nominal data alone, with **ADWIN concept-drift diagnosis** for **early identification of emerging shift**.
- Cut **false alarms from 49.87% to under 1%** while sustaining **F1 up to 0.80**, the failure mode that makes naive monitoring unusable.
- Reached **88.4% data reduction** and **3.8 ms latency**; deployed containerized services with **continuous monitoring** through InfluxDB and Grafana across **48 controlled experiments**.

ACHIEVEMENTS

WayneHacks 4 Hackathon – 3rd Place of 30+ Teams, Wayne State University and TechTown Detroit

Mar 2026

Aligned a **3-person cross-functional team** under a 24-hour constraint and owned the model pipeline for an elderly-fall safety monitor at **75.6% accuracy**.